

LAB 1

REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

Problem Statement #64

DISASTER RELIEF SUPPLY & VOLUNTEER COORDINATOR

Domain: Sustainability & Green Tech

NAME: SAHANA K GOLABANVI

SRN: PES1UG25AM811

SECTION: D

Actors: Relief Volunteer, Disaster Manager

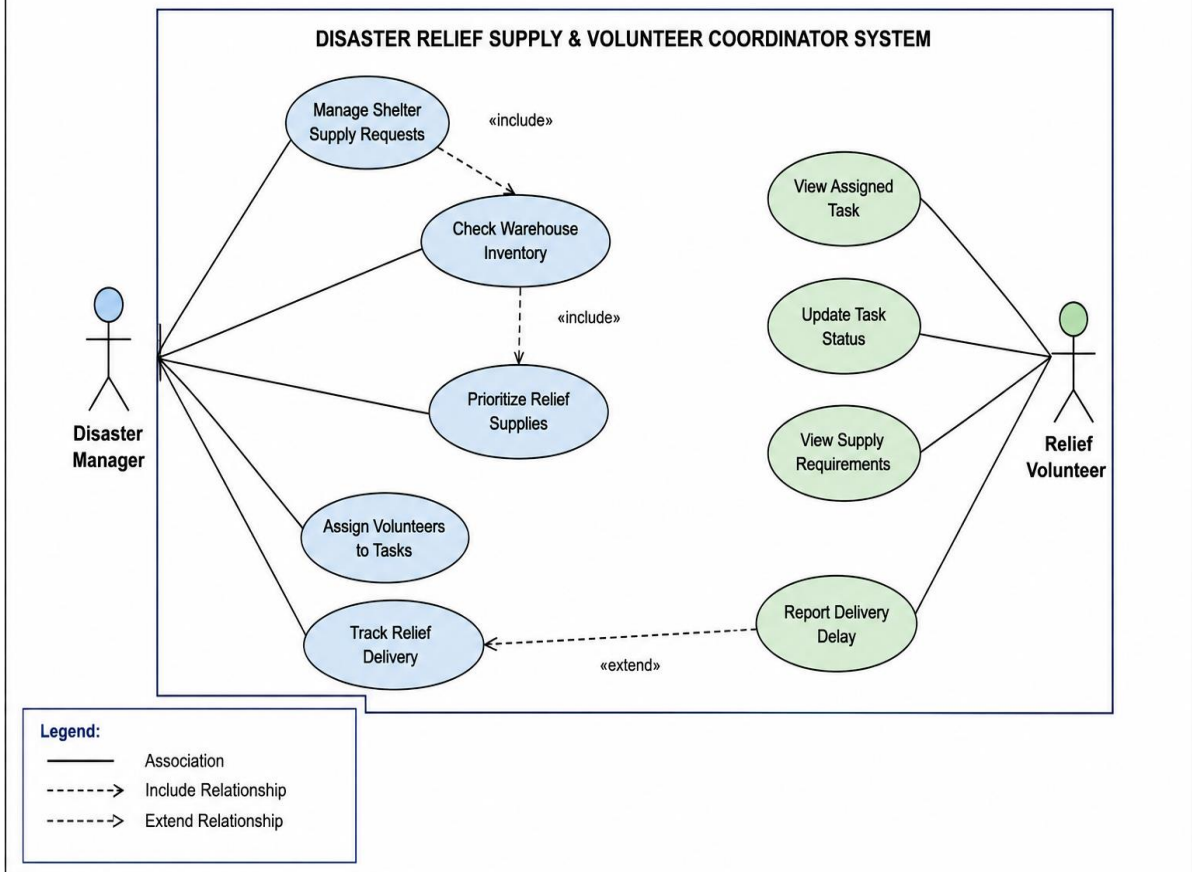
ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall record and manage incoming shelter supply requisitions for critical items such as water, blankets, and medical kits.	High	Pass: The system correctly records the shelter, requested item, quantity, and urgency. Fail: A valid requisition is missing or incorrectly recorded.	To maintain accurate records of shelter supply needs during disaster relief operations.
FR-002	Functional	The system shall match incoming shelter supply requisitions against available central warehouse supplies.	High	Pass: Available warehouse quantities are correctly matched with the requested supplies. Fail: Available stock is ignored or incorrectly matched.	To ensure available relief supplies are efficiently allocated to shelters.
FR-003	Functional	The system shall assign delivery priority based on the criticality of the requested shelter supplies.	High	Pass: Critical needs such as water and medical kits receive higher dispatch priority. Fail: A lower-priority item is dispatched ahead of a critical need.	To ensure critical disaster-relief needs are addressed before less urgent requests.

FR-004	Functional	The system shall match available volunteers with relief tasks based on their recorded skill sets and availability.	High	Pass: Suitable volunteers are identified for the required relief task. Fail: Volunteers with unsuitable skills are assigned when suitable volunteers are available.	To deploy volunteers effectively according to their skills and availability.
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ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-005	Functional	The system shall track relief vehicle deliveries and update the delivery status.	High	Pass: The delivery status is updated correctly from dispatch through completion. Fail: A delivery status is missing or not updated correctly.	To provide visibility of relief-supply deliveries and their current status.
NFR-001	Performance & Security	The disaster mapping dashboard shall operate with low network bandwidth consumption and support offline GIS data caching.	High	Pass: The dashboard remains usable under low-bandwidth conditions and cached GIS data is available offline. Fail: The dashboard becomes unusable or required cached GIS data cannot be accessed offline.	To support disaster-management operations in areas with unreliable or limited network connectivity.
NFR-002	Reliability	The system shall maintain accurate shelter requests, warehouse inventory, volunteer assignments, and delivery-status information during normal operation.	High	Pass: Stored operational information remains accurate after system operations. Fail: Information is lost, duplicated, or incorrectly updated.	To ensure dependable coordination of relief supplies, volunteers, and deliveries.

2. USE CASE DIAGRAM:

2. UML USE-CASE DIAGRAM



USE-CASE FLOW SPECIFICATION

Disaster Relief Supply & Volunteer Coordinator

Use Case: Match and Dispatch Relief Supplies

Primary Actor: Disaster Manager

Supporting Actor: Relief Volunteer

1. Preconditions

1. The Disaster Manager is logged into the system.
2. A shelter supply requisition is available in the system.
3. Central warehouse inventory information is available.
4. Relief-supply priority rules are configured or available.

2. Postconditions

1. The shelter supply requisition is matched against available warehouse supplies.
2. The required delivery priority is assigned.
3. Dispatch information is recorded for tracking.
4. The relevant relief task can be assigned to an available volunteer when required.

3. Main Success Scenario

1. The Disaster Manager logs into the system.
2. The system displays incoming shelter supply requisitions.
3. The Disaster Manager selects a shelter supply requisition.
4. The system checks the central warehouse inventory.
5. The system matches requested items with available quantities.
6. The system determines the delivery priority based on the criticality of the request.
7. The system prepares the dispatch information for the required supplies.
8. The Disaster Manager confirms the dispatch.
9. The system records the dispatch and delivery-tracking information.
10. The system can identify suitable Relief Volunteers for related relief tasks based on their skill sets and availability.

4. Alternate Flow

A1. Required Supply Is Not Available in the Warehouse

1. The system checks the warehouse inventory for all requested items.
2. One or more requested items are found to be unavailable or insufficient.
3. The system marks the unavailable quantity and informs the Disaster Manager.
4. The system does not dispatch the unavailable quantity as if it were available.

5. The Disaster Manager can arrange an alternate source or update the requisition.

5. Expected Outcome

The system successfully matches shelter supply requisitions with available central warehouse supplies, assigns appropriate delivery priority, records dispatch information, and supports volunteer coordination for relief tasks.

6. Actor-to-Use-Case Summary

Actor	Associated Use Cases
Disaster Manager	Manage Shelter Supply Requests; Check Warehouse Inventory; Prioritize Relief Supplies; Assign Volunteers; Track Relief Delivery
Relief Volunteer	View Assigned Task; Update Task Status; View Supply Requirements; Report Delivery Delay

7. Key Relationships Used in UML Diagram

«include»: Manage Shelter Supply Requests includes Check Warehouse Inventory because inventory availability is required to process the supply request.

«include»: Check Warehouse Inventory includes Prioritize Relief Supplies because priority is determined as part of dispatch planning.

«extend»: Report Delivery Delay extends Track Relief Delivery because reporting a delay occurs only when a delivery problem is encountered.

8. Scenario Reference

Problem Statement #64 – Sustainability & Green Tech: Disaster Relief Supply & Volunteer Coordinator.

Target Actors: Relief Volunteer and Disaster Manager.